Srushti Nerkar

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Education

•Malaviya National Institute of Technology (MNIT), Jaipur *2022 – 2026*
B.Tech, Computer Science and Engineering

Technical Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL
Backend: Spring Boot, Spring Data JPA, Node.js, Express.js, Flask, REST APIs, Microservices, Prisma ORM, Drizzle ORM
Frontend: React, TypeScript, Tailwind CSS, Vite
Databases & Messaging: PostgreSQL (pgvector), MySQL, MongoDB, Redis, Kafka
DevOps & Tools: Docker, Git, Linux, CI/CD, Render, Neon, Firebase Auth, Clerk
AI/ML: PyTorch, Scikit-learn, RAG, LLM Tool-Calling, Vector Search, Spiking Neural Networks
Fundamentals: Data Structures and Algorithms, OOP, LLD (SOLID, Design Patterns), HLD, Operating Systems, Computer Networks (TCP/IP)

Experience

•Birdvision Consulting Pvt. Ltd. *May 2025 – July 2025*
Software Development Intern Pune
– Architected Flask REST microservices for messaging, routing and state workflows in a WhatsApp Business platform serving 10,000+ active users; decomposed by domain boundary into independently deployable services, cutting cross-service dependencies 35%.
– Re-engineered centralized error-handling middleware and response-parsing pipelines across the API layer, cutting unhandled exceptions 40%.

•Google Developer Student Clubs (GDSC) *Oct 2023 – Sep 2024*
Technical Executive Jaipur
– Led a hands-on machine learning workshop for 150+ engineering students; designed the lab modules and published reference code repositories.

Projects

•AeroPath: Connected Travel Timeline Engine *Oct 2026 – Present*
React 19, TypeScript, Node.js, Express, PostgreSQL (pgvector), Prisma, Spring Boot, Docker GitHub
– Built flight and hotel search workflows over live third-party REST APIs (AeroDataBox, Hotels4) with backend-driven caching, cutting external API response latency by 45%.
– Implemented atomic Firebase-to-PostgreSQL user synchronization using Firebase Admin SDK and Prisma ORM in Node.js.
– Extended the system into a polyglot microservice architecture with a Spring Boot (Spring Data JPA, Docker) service for real-time flight disruption alerting.
– In progress: LLM tool-calling for conversational itinerary generation and a pgvector-backed RAG semantic search engine.

•Job Watcher: Fresher Job Tracker *Oct 2026*
React, TypeScript, Node.js, Express, PostgreSQL (Drizzle), Clerk, Nodemailer, Render, Neon Live
– Built a cron-driven job aggregator that monitors enterprise career feeds, parses listings, and sends real-time notifications to Clerk-authenticated users via Nodemailer.
– Designed indexed PostgreSQL schemas with Drizzle ORM, enabling sub-50ms multi-parameter filtering by experience level, tech stack, and geolocation.

•TraceFlow: Live C++ Code-to-Flowchart Visualiser *Sep 2026 – Oct 2026*
React, TypeScript, Vite, Monaco, Tree-sitter (WASM), React Flow, dagre Live
– Built a fully client-side static analysis tool (no backend) that converts C++ source into interactive Control Flow Graphs using Tree-sitter WASM parsers and dagre layout.
– Added shareable permalinks, PNG/SVG graph export, and side-by-side complexity comparison of brute-force vs. optimized solutions.

Patents, Research & Achievements

•Granted 4 Indian Patents for innovations in distributed routing and systems *2026*
•Co-authored "PRISM: Progressive Routing for Intrusion Detection via Spiking Multi-layer Networks", submitted to ICMLDE 2026 *2026*
•Finalist at Baker Hughes Hackathon 2025 *2025*

Competitive Programming

- **LeetCode (Max Rating 2199, Top 0.93%)**: Global Rank 140 in Biweekly Contest 193 | Link
- **GeeksforGeeks (Max Rating 1756, 3-star)**: Global Rank 210 in Weekly 189 | Link
- **CodeChef (Max Rating 1725, 3-star)**: Global Rank 180 in Starters 257 | Link